

Please check the examination details below before entering your candidate information	
Candidate surname	Other names
Centre Number	Candidate Number
Pearson Edexcel Level 3 GCE	
Friday 13 June 2025	
Afternoon (Time: 1 hour 30 minutes)	Paper reference 9FM0/3B
Further Mathematics Advanced PAPER 3B: Further Statistics 1	
You must have: Mathematical Formulae and Statistical Tables (Green), calculator	Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of the tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
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Turn over ►

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1. Irina is practising her serves in badminton and counts the number of her serves that are faults. She finds that 15% of her serves are faults.

Assuming that each serve is independent,

- (a) find the probability that

- (i) Irina's 4th fault comes on her 20th serve,

(2)

- (ii) in 18 serves, Irina has 4 faults.

(2)

With practice, Irina reduces her proportion of faults, p , so that the mean number of serves until her 4th fault is at least 32

- (b) Find the maximum value of p

(3)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 7 marks)



2. The discrete random variable X has probability distribution

x	-1	a	b
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$

where a and b are positive constants.

(a) Find an expression for $E(X)$ in terms of a and b

(2)

The discrete random variable Y is defined as $Y = a + bX$

Given that $\text{Var}(Y) = \frac{1}{4} \text{Var}(X)$ and $E(Y) = \frac{5}{16}$

(b) find the value of $E(X)$

(7)



DO NOT WRITE IN THIS AREA

Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 9 marks)



3. A journalist uses a gender equality test on some randomly chosen films. She believes that whether or not a film passes the test is associated with the period in which the film is first released. Her data is summarised in Table 1. The journalist decides to test whether the data supports her belief.

Table 1: Observed frequencies

Test result \ Period	1990–1999	2000–2004	2005–2013	Total
Pass	45	70	65	180
Fail	75	90	55	220
Total	120	160	120	400

Some of the expected frequencies she calculates are shown in Table 2 on page 7.

- (a) Complete Table 2 showing expected frequencies. (2)
- (b) Write down hypotheses for a suitable test to assess the journalist's belief. (1)
- (c) Showing your working clearly, test the journalist's belief using a 5% level of significance. (6)
- You should state your critical value and conclusion clearly.

The journalist wants to publish her data and findings. She decides to publish the data as **percentages** of pass and fail for each period, as shown in Table 3, rather than as **frequencies**.

Table 3: Percentages

Test result \ Period	1990–1999	2000–2004	2005–2013
Pass (%)	37.5	43.75	54.17
Fail (%)	62.5	56.25	45.83
Total (%)	100	100	100

The editor suggests carrying out the hypothesis test of the journalist's belief using the percentages in Table 3.

- (d) (i) Describe the effect this would have on the test statistic, justifying your answer. (3)
- (ii) Explain whether or not the journalist should follow her editor's suggestion.



Question 3 continued

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DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 12 marks)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 10 marks)



Given that in their next match the team scores exactly 3 goals,

A hockey team plays 2 games each week during a 40-week season.

Each game consists of 2 periods.

The probability that the team concedes no goals in a period is 0.55

The random variable X represents the number of **periods** in a week in which the team concedes no goals.

(c) (i) Write down a suitable distribution for X

(ii) For this 40-week season, use the Central Limit Theorem to estimate $P(\bar{X} > 2)$

(4)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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DO NOT WRITE IN THIS AREA

Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 10 marks)



6. A biased coin and a fair coin are thrown repeatedly.

The probability that the biased coin shows heads is $\frac{1}{3}$

Let B represent the number of throws of the biased coin until it shows heads for the first time. Let F represent the number of throws of the fair coin until it shows heads for the first time.

(a) Find

$$(i) \quad P(B = 3) \tag{2}$$

$$(ii) \quad P(2 \leq F \leq 8) \tag{3}$$

$$(iii) \quad P(\{B=3\} \cup \{F=3\}) \tag{3}$$

Each day Chris invites his friend Shivani to play a game with these coins. One player's score will be X and the other player's score will be Y , where

$$X=4F \quad \text{and} \quad Y=\frac{B^2}{2}$$

Chris and Shivani will play a large number of games and the winner will be the player with the higher **total** score.

Before their first game, Chris allows Shivani to choose whether her score for every game will be X or her score for every game will be Y

- (b) State, explaining your reasoning and showing any calculations you have made, which choice Shivani should make.
- (5)**



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 13 marks)



7. A software company develops computer programs.

Its *Lovelace* program is found to glitch 7 times on average when it is run.

The company makes some alterations to improve the program.

It then tests to see if it has been successful in reducing the average number of glitches per run.

The glitches can be assumed to occur independently.

- (a) (i) State suitable hypotheses for the test.

(ii) Find the critical region for the test at a 10% level of significance.

(iii) State the size of the test.

(4)

The altered program is actually found to glitch 5 times on average when run.

- (b) Find the probability of a Type II error for the company's test.

(3)

Due to the high value of the probability of a Type II error in part (b), the company makes an entirely new version of the program.

The new version of the program is designed to have an average of fewer than k glitches per run.

The random variable T represents the number of **runs** of the program until a run with no glitches occurs.

- (c) Write down the name of a suitable distribution for T

(1)

If the new version has an average of λ glitches per run, this new version of the program will be tested using the hypotheses

$$H_0: \lambda = k$$

$$H_1: \lambda < k$$

The company will reject H_0 if the first run without any glitches of the new version occurs within n runs.

- (d) Show that the power function of the test is given by

$$1 - (1 - e^{-\lambda})^n$$

(2)

The company requires $P(\text{Type II error}) < 0.2$

Assuming that the new version of the program has an average of 2.75 glitches per run,

- (e) find the minimum value of n

(4)



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 14 marks)

TOTAL FOR PAPER IS 75 MARKS



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Further Mathematics Advanced PAPER 3B: Further Statistics 1	
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Turn over ►

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1. The discrete random variable X has the following probability distribution

x	-1	0	1	3	5
$P(X=x)$	0.2	0.1	0.2	0.25	0.25

- (a) Find $\text{Var}(X)$

(3)

- (b) Find $\text{Var}(X^2)$

(3)



DO NOT WRITE IN THIS AREA

Question 1 continued

Handwriting practice area with horizontal lines.

(Total for Question 1 is 6 marks)



2. The number of errors made by a secretary is modelled by a Poisson distribution with a mean of 2.4 per 100 words.

A 100-word piece of work completed by the secretary is selected at random.

- (a) Find the probability that

- (i) there are exactly 3 errors,
(ii) there are fewer than 2 errors.

(2)

After a long holiday, a randomly selected piece of work containing 250 words completed by the secretary is examined to see if the rate of errors has changed.

- (b) Stating your hypotheses clearly, and using a 5% level of significance, find the critical region for a suitable test.

(4)

- (c) Find $P(\text{Type I error})$ for the test in part (b)

(1)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



3. Tisam took a survey of students' favourite colours. The results are summarised in the table below.

		Colour					
		Red	Blue	Green	Yellow	Black	Total
Year group	1–5	34	15	14	22	3	88
	6–9	23	32	12	9	8	84
	10–12	5	28	19	8	8	68
	Total	62	75	45	39	19	240

Tisam carries out a suitable test to see if there is any association between favourite colour and year group.

- (a) Write down the hypotheses for a suitable test.

(1)

For her table, Tisam only needs to check one cell to show that none of the expected frequencies are less than 5

- (b) (i) Identify this cell, giving your reason.

- (ii) Calculate the expected frequency for this cell.

(2)

The test statistic for Tisam's test is 38.449

- (c) Using a 1% level of significance, complete the test.
You should state your critical value and conclusion clearly.

(3)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 6 marks)



4. Every morning Geethaka repeatedly rolls a fair, six-sided die until he rolls a 3 and then he stops. The random variable X represents the number of times he rolls the die each morning.

(a) Suggest a suitable model for the random variable X (1)

(b) Show that $P(X \leq 3) = \frac{91}{216}$ (2)

After 64 mornings Geethaka will calculate the mean number of times he rolled the die.

(c) Estimate the probability that the mean number of rolls is between 5.6 and 7.2 (5)

Nira wants to check Geethaka's die to decide whether or not the probability of rolling a 3 with his die is less than $\frac{1}{6}$

Nira rolls the die repeatedly until she rolls a 3
She obtains $x = 16$

(d) By carrying out a suitable test, determine what Nira's conclusion should be. You should state your hypotheses clearly and use a 5% level of significance. (4)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 12 marks)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 10 marks)



- (2)

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 16 marks)



7. The probability of winning a prize when playing a single game of *Pento* is $\frac{1}{5}$

When more than one game is played the games are independent.
Sam plays 20 games.

- (a) Find the probability that Sam wins 4 or more prizes. (2)

Tessa plays a series of games.

- (b) Find the probability that Tessa wins her 4th prize on her 20th game. (2)

Rama invites Sam and Tessa to play some new games of *Pento*.

They must pay Rama £1 for each game they play but Rama will pay them £2 for the first time they win a prize, £4 for the second time and £(2w) when they win their wth prize ($w > 2$)

Sam decides to play n games of *Pento* with Rama.

- (c) Show that Sam's expected profit is $\pounds \frac{1}{25}(n^2 - 16n)$ (6)

Given that Sam chose $n = 15$

- (d) find the probability that Sam does not make a loss. (4)

Tessa agrees to play *Pento* with Rama. She will play games until she wins r prizes and then she will stop.

- (e) Find, in terms of r , Tessa's expected profit. (4)



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 18 marks)

TOTAL FOR PAPER IS 75 MARKS



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Pearson Edexcel Level 3 GCE

Friday 16 June 2023

Afternoon (Time: 1 hour 30 minutes) **Paper reference** **9FM0/3B**

Further Mathematics

Advanced

PAPER 3B: Further Statistics 1

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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1. The discrete random variable X has probability distribution

x	-2	-1	0	1	3
$P(X=x)$	0.25	a	b	a	0.30

where a and b are probabilities.

- (a) Find $E(X)$

(2)

Given that $\text{Var}(X) = 3.9$

- (b) find the value of a and the value of b

(4)

The independent random variables X_1 and X_2 each have the same distribution as X

- (c) Find $P(X_1 + X_2 > 3)$

(3)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 9 marks)



2. Telephone calls arrive at a call centre randomly, at an average rate of 1.7 per minute. After the call centre was closed for a week, in a random sample of 10 minutes there were 25 calls to the call centre.

- (a) Carry out a suitable test to determine whether or not there is evidence that the rate of calls arriving at the call centre has changed.

Use a 5% level of significance and state your hypotheses clearly.

(4)

Only 1.2% of the calls to the call centre last longer than 8 minutes.

One day Tiang has 70 calls.

- (b) Find the probability that out of these 70 calls Tiang has more than 2 calls lasting longer than 8 minutes.

(3)

The call centre records show that 95% of days have at least one call lasting longer than 30 minutes.

On Wednesday 900 calls arrived at the call centre and none of them lasted longer than 30 minutes.

- (c) Use a Poisson approximation to estimate the proportion of calls arriving at the call centre that last longer than 30 minutes.

(4)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 11 marks)



3. In a class experiment, each day for 170 days, a child is chosen at random and spins a large cardboard coin 5 times and the number of heads is recorded. The results are summarised in the following table.

Number of heads	0	1	2	3	4	5
Frequency	3	10	45	62	38	12

Marcus believes that a $B(5, 0.5)$ distribution can be used to model these data and he calculates expected frequencies, to 2 decimal places, as follows

Number of heads	0	1	2	3	4	5
Expected frequency	r	26.56	s	s	26.56	r

- (a) Find the value of r and the value of s (3)
- (b) Carry out a suitable test, at the 5% level of significance, to determine whether or not the $B(5, 0.5)$ distribution is a good model for these data. You should state clearly your hypotheses, the test statistic and the critical value used. (6)

Nima believes that a better model for these data would be $B(5, p)$

- (c) Find a suitable estimate for p (1)

To test her model, Nima uses this value of p , to calculate expected frequencies as follows

Number of heads	0	1	2	3	4	5
Expected frequency	2.07	14.65	41.44	58.63	41.47	11.74

The test statistic for Nima's test is 1.62 (to 3 significant figures)

- (d) State,
 (i) giving your reasons, the degrees of freedom
 (ii) the critical value
 that Nima should use for a test at the 5% significance level. (3)
- (e) With reference to Marcus' and Nima's test results, comment on
 (i) the probability of the coin landing on heads,
 (ii) the independence of the spins of the coin.
 Give reasons for your answers. (2)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 15 marks)



4. There are 32 students in a class.

Each student rolls a fair die repeatedly, stopping when their total number of sixes is 4
Each student records the total number of times they rolled the die.

Estimate the probability that the mean number of rolls for the class is less than 27.2

(6)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 6 marks)



5. A machine fills cartons with juice.

The amount of juice in a carton is normally distributed with mean μ ml and standard deviation 8 ml.

A manager wants to test whether or not the amount of juice in the cartons, X ml, is less than 330 ml. The manager takes a random sample of 25 cartons of juice and calculates the mean amount of juice $\bar{x}$ ml.

- (a) Using a 5% level of significance, find the critical region of $\bar{X}$ for this test. State your hypotheses clearly.

(4)

The Director is concerned about the machine filling the cartons with more than 330 ml of juice as well as less than 330 ml of juice. The Director takes a sample of 55 cartons, records the mean amount of juice $\bar{y}$ ml and uses a test with a critical region of

$$\{\bar{Y} < 328\} \cup \{\bar{Y} > 332\}$$

- (b) Find $P(\text{Type I error})$ for the Director's test.

(2)

When $\mu = 325$ ml

- (c) find $P(\text{Type II error})$ for the test in part (a)

(2)



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

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(Total for Question 5 is 8 marks)



- $$G_X(t) = \frac{t^2}{(3-2t)^2}$$

- A fair die is rolled repeatedly.

- (c) Use calculus and $G_X(t)$ to find

- $$\begin{aligned} \text{(i)} \quad & E(X) \\ \text{(ii)} \quad & \text{Var}(X) \end{aligned} \tag{7}$$

The discrete random variable Y has probability generating function

$$G_Y(t) = \frac{t^{10}}{(3 - 2t^3)^2}$$

- (d) Find the exact value of $P(Y = 19)$ (3)

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 13 marks)



DO NOT WRITE IN THIS AREA

Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 13 marks)

TOTAL FOR PAPER IS 75 MARKS



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Pearson Edexcel Level 3 GCE

Time 1 hour 30 minutes

Paper
reference

9FM0/3B

Further Mathematics

Advanced

PAPER 3B: Further Statistics 1

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Turn over ►

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1. A researcher is investigating the number of female cubs present in litters of size 4. He believes that the number of female cubs in a litter can be modelled by $B(4, 0.5)$. He randomly selects 100 litters each of size 4 and records the number of female cubs. The results are recorded in the table below.

Number of female cubs	0	1	2	3	4
Observed number of litters	10	33	33	15	9

He calculated the expected frequencies as follows

Number of female cubs	0	1	2	3	4
Expected number of litters	6.25	r	s	r	6.25

- (a) Find the value of r and the value of s (3)
- (b) Carry out a suitable test, at the 5% level of significance, to determine whether or not the number of female cubs in a litter can be modelled by $B(4, 0.5)$. You should clearly state your hypotheses and the critical value used. (6)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 9 marks)



2. The discrete random variable X has probability distribution

x	-5	-1	0	5	b
$P(X=x)$	0.3	0.25	0.1	0.15	0.2

where b is a constant and $b > 5$

(a) Find $E(X)$ in terms of b

(1)

Given that $\text{Var}(X) = 34.26$

(b) find the value of b

(4)

(c) Find $P(X^2 < 2 - 3X)$

(4)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 9 marks)



3. During the summer, mountain rescue team A receives calls for help randomly with a rate of 0.4 per day.

- (a) Find the probability that during the summer, mountain rescue team A receives at least 19 calls for help in 28 randomly selected days. (2)

The leader of mountain rescue team A randomly selects 250 summer days from the last few years.

She records the number of calls for help received on each of these days.

- (b) Using a Poisson approximation, estimate the probability of the leader finding at least 20 of these days when more than 1 call for help was received by mountain rescue team A . (4)

Mountain rescue team A believes that the number of calls for help per day is lower in the winter than in the summer. The number of calls for help received in 42 randomly selected winter days is 8

- (c) Use a suitable test, at the 5% level of significance, to assess whether or not there is evidence that the number of calls for help per day is lower in the winter than in the summer. State your hypotheses clearly. (4)

During the summer, mountain rescue team B receives calls for help randomly with a rate of 0.2 per day, independently of calls to mountain rescue team A .

The random variable C is the total number of calls for help received by mountain rescue teams A and B during a period of n days in the summer.

On a Monday in the summer, mountain rescue teams A and B each receive a call for help.

Given that over the next n days $P(C = 0) < 0.001$

- (d) calculate the minimum value of n (3)
- (e) Write down an assumption that needs to be made for the model to be appropriate. (1)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 14 marks)



4. In a game a spinner is spun repeatedly. When the spinner is spun, the probability of it landing on blue is 0.11

(a) Find the probability that the spinner lands on blue

(i) for the first time on the 6th spin,

(2)

(ii) for the first time before the 6th spin,

(2)

(iii) exactly 4 times during the first 6 spins,

(2)

(iv) for the 4th time on or before the 6th spin. (4)

Zac and Izana play the game. They take turns to spin the spinner. The winner is the first one to have the spinner land on blue. Izana spins the spinner first.

(b) Show that the probability of Zac winning is 0.471 to 3 significant figures. (3)

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Question 4 continued

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Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 13 marks)



5. A random sample of 150 observations is taken from a geometric distribution with parameter 0.3

Estimate the probability that the mean of the sample is less than 3.45

(5)

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Question 5 continued

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(Total for Question 5 is 5 marks)



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- 6.** The discrete random variable V has probability distribution

v	2	3	4
$P(V = v)$	$\frac{9}{25}$	$\frac{12}{25}$	$\frac{4}{25}$

- (a) Show that the probability generating function of V is

$$G_V(t) = t^2 \left(\frac{2}{5}t + \frac{3}{5} \right)^2 \quad (2)$$

The discrete random variable W has probability generating function

$$G_W(t) = t \left(\frac{2}{5}t + \frac{3}{5} \right)^5$$

- (b) Use calculus to find

- $$(i) \ E(W) \tag{2}$$

- $$\text{(ii) } \text{Var}(W) \tag{4}$$

Given that V and W are independent,

- (c) find the probability generating function of $X = V + W$ in its simplest form. (2)

The discrete random variable $Y = 2X + 3$

- (d) Find the probability generating function of Y (2)

- (e) Find $P(Y = 15)$ (2)



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Question 6 continued

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Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 14 marks)



7. A machine fills bags with flour. The weight of flour delivered by the machine into a bag, X grams, is normally distributed with mean μ grams and standard deviation 30 grams. To check if there is any change to the mean weight of flour delivered by the machine into each bag, Olaf takes a random sample of 10 bags. The weight of flour, x grams, in each bag is recorded and $\bar{x} = 1020$

(a) Test, at the 5% level of significance, $H_0: \mu = 1000$ against $H_1: \mu \neq 1000$

(4)

Olaf decides to alter the test so that the hypotheses are $H_0: \mu = 1000$ and $H_1: \mu > 1000$ but keeps the level of significance at 5%

He takes a second sample of size n and finds the critical region, $\bar{X} > c$

(b) Find an equation for c in terms of n

(2)

When the true value of μ is 1020 grams, the probability of making a Type II error is 0.0050, to 2 significant figures.

(c) Calculate the value of n and the value of c

(5)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 11 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Time 1 hour 30 minutes	Paper reference	9FM0/3B	
Further Mathematics Advanced PAPER 3B: Further Statistics 1			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
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- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
- Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of the tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over ►

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1. Kelly throws a tetrahedral die n times and records the number on which it lands for each throw.

She calculates the expected frequency for each number to be 43 if the die was unbiased.

The table below shows three of the frequencies Kelly records but the fourth one is missing.

Number	1	2	3	4
Frequency	47	34	36	x

- (a) Show that $x = 55$

(1)

Kelly wishes to test, at the 5% level of significance, whether or not there is evidence that the tetrahedral die is unbiased.

- (b) Explain why there are 3 degrees of freedom for this test.

(1)

- (c) Stating your hypotheses clearly and the critical value used, carry out the test.

(5)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 7 marks)



2. On a weekday, a garage receives telephone calls randomly, at a mean rate of 1.25 per 10 minutes.
- (a) Show that the probability that on a weekday at least 2 calls are received by the garage in a 30-minute period is 0.888 to 3 decimal places. (2)
- (b) Calculate the probability that at least 2 calls are received by the garage in fewer than 4 out of 6 randomly selected, non-overlapping 30-minute periods on a weekday. (2)

The manager of the garage randomly selects 150 non-overlapping 30-minute periods on weekdays.

She records the number of calls received in each of these 30-minute periods.

- (c) Using a Poisson approximation show that the probability of the manager finding at least 3 of these 30-minute periods when exactly 8 calls are received by the garage is 0.664 to 3 significant figures. (4)
- (d) Explain why the Poisson approximation may be reasonable in this case. (1)

The manager of the garage decides to test whether the number of calls received on a Saturday is different from the number of calls received on a weekday. She selects a Saturday at random and records the number of telephone calls received by the garage in the first 4 hours.

- (e) Write down the hypotheses for this test. (1)

The manager found that there had been 40 telephone calls received by the garage in the first 4 hours.

- (f) Carry out the test using a 5% level of significance. (4)



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Question 2 continued

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Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 14 marks)



3. A courier delivers parcels. The random variable X represents the number of parcels delivered successfully each day by the courier where $X \sim B(400, 0.64)$

A random sample $X_1, X_2, \dots, X_{100}$ is taken.

Estimate the probability that the mean number of parcels delivered each day by the courier is greater than 257

(4)



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Question 3 continued

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(Total for Question 3 is 4 marks)



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Question 4 continued

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Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 10 marks)



5. Asha, Davinda and Jerry each have a bag containing a large number of counters, some of which are white and the rest are red.
Each person draws counters from their bag one at a time, notes the colour of the counter and returns it to their bag.

The probability of Asha getting a red counter on any one draw is 0.07

- (a) Find the probability that Asha will draw at least 3 white counters before a red counter is drawn. (2)
- (b) Find the probability that Asha gets a red counter for the second time on her 9th draw. (2)

The probability of Davinda getting a red counter on any one draw is p .
Davinda draws counters until she gets n red counters. The random variable D is the number of counters Davinda draws.

Given that the mean and the standard deviation of D are 4400 and 660 respectively,

- (c) find the value of p . (4)

Jerry believes that his bag contains a smaller proportion of red counters than Asha's bag. To test his belief, Jerry draws counters from his bag until he gets a red counter. Jerry defines the random variable J to be the number of counters drawn up to and including the first red counter.

- (d) Stating your hypotheses clearly and using a 10% level of significance, find the critical region for this test. (5)

Jerry gets a red counter for the first time on his 34th draw.

- (e) Giving a reason for your answer, state whether or not there is evidence that Jerry's bag contains a smaller proportion of red counters than Asha's bag. (2)

Given that the probability of Jerry getting a red counter on any one draw is 0.011

- (f) show that the power of the test is 0.702 to 3 significant figures. (3)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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6. The probability generating function of the random variable X is

$$G_X(t) = k(1 + 2t)^5$$

where k is a constant.

- (a) Show that $k = \frac{1}{243}$ (2)
- (b) Find $P(X = 2)$ (2)
- (c) Find the probability generating function of $W = 2X + 3$ (2)

The probability generating function of the random variable Y is

$$G_Y(t) = \frac{t(1 + 2t)^2}{9}$$

Given that X and Y are independent,

- (d) find the probability generating function of $U = X + Y$ in its simplest form. (2)
- (e) Use calculus to find the value of $\text{Var}(U)$ (6)



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Question 6 continued

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Question 6 continued

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Question 6 continued

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(Total for Question 6 is 14 marks)



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Question 7 continued

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Question 7 continued

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(Total for Question 7 is 8 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Friday 16 October 2020			
Afternoon (Time: 1 hour 30 minutes)		Paper Reference 9FM0/3B	
Further Mathematics Advanced Paper 3B: Further Statistics 1			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- Answer the questions in the spaces provided
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- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of the tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Question 1 continued

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Question 1 continued

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Question 1 continued

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(Total for Question 1 is 13 marks)



2. The discrete random variables W , X and Y are distributed as follows

$$W \sim B(10, 0.4) \quad X \sim \text{Po}(4) \quad Y \sim \text{Po}(3)$$

(a) Explain whether or not $\text{Po}(4)$ would be a good approximation to $B(10, 0.4)$ (1)

(b) State the assumption required for $X + Y$ to be distributed as $\text{Po}(7)$ (1)

Given the assumption in part (b) holds,

(c) find $P(X + Y < \text{Var}(W))$ (2)



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Question 2 continued

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(Total for Question 2 is 4 marks)



3. Suzanne and Jon are playing a game.

They put 4 red counters and 1 blue counter in a bag.

Suzanne reaches into the bag and selects one of the counters at random. If the counter she selects is blue, she wins the game. Otherwise she puts it back in the bag and Jon selects one at random. If the counter he selects is blue, he wins the game. Otherwise he puts it back in the bag and they repeat this process until one of them selects the blue counter.

- (a) Find the probability that Suzanne selects the blue counter on her 4th selection. (2)
- (b) Find the probability that the blue counter is first selected on or after Jon's third selection. (2)
- (c) Find the mean and standard deviation of the number of selections made until the blue counter is selected. (2)
- (d) Find the probability that Suzanne wins the game. (3)



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Question 3 continued

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Question 3 continued

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Question 3 continued

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(Total for Question 3 is 9 marks)



4. The discrete random variable X has the following probability distribution.

x	-5	-2	3	4
$P(X = x)$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{2}$

- (a) Find $\text{Var}(X)$

(3)

The discrete random variable Y is defined in terms of the discrete random variable X

When X is negative, $Y = X^2$

When X is positive, $Y = 3X - 2$

- (b) Find $P(Y < 9)$

(3)

- (c) Find $E(XY)$

(2)



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Question 4 continued

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Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 8 marks)



5. A factory produces pins.

An engineer selects 40 independent random samples of 6 pins produced at the factory and records the number of defective pins in each sample.

Number of defective pins	0	1	2	3	4	5	6
Observed frequency	19	11	7	2	0	1	0

(a) Show that the proportion of defective pins in the 40 samples is 0.15

(2)

The engineer suggests that the number of defective pins in a sample of 6 can be modelled using a binomial distribution. Using the information from the sample above, a test is to be carried out at the 10% significance level, to see whether the data are consistent with the engineer's suggested model.

The value of the test statistic for this test is 2.689

(b) Justifying the degrees of freedom used, carry out the test, at the 10% significance level, to see whether the data are consistent with the engineer's suggested model. State your hypotheses clearly.

(8)

The engineer later discovers that the previously recorded information was incorrect. The data should have been as follows.

Number of defective pins	0	1	2	3	4	5	6
Observed frequency	19	11	6	3	1	0	0

(c) Describe the effect this would have on the value of the test statistic that should be used for the hypothesis test. Give reasons for your answer.

(3)



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Question 5 continued

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Question 5 continued

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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 13 marks)



6. A discrete random variable X has probability generating function given by

$$G_X(t) = \frac{1}{64} (a + bt^2)^2$$

where a and b are positive constants.

- (a) Write down the value of $P(X = 3)$

(1)

Given that $P(X = 4) = \frac{25}{64}$

- (b) (i) find $P(X = 2)$

(7)

- (ii) find $E(X)$

(3)

The random variable $Y = 3X + 2$

- (c) Find the probability generating function of Y

(2)



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Question 6 continued

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Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 13 marks)



7. A six-sided die has sides labelled 1, 2, 3, 4, 5 and 6

The random variable S represents the score when the die is rolled.

Alicia rolls the die 45 times and the mean score, $\bar{S}$, is calculated.

Assuming the die is fair and using a suitable approximation,

- (a) find, to 3 significant figures, the value of k such that $P(\bar{S} < k) = 0.05$ (8)

- (b) Explain the relevance of the Central Limit Theorem in part (a). (2)

Alicia considers the following hypotheses:

H_0 : The die is fair

H_1 : The die is not fair

If $\bar{S} < 3.1$ or $\bar{S} > 3.9$, then H_0 will be rejected.

Given that the true distribution of S has mean 4 and variance 3

- (c) find the power of this test. (3)

- (d) Describe what would happen to the power of this test if Alicia were to increase the number of rolls of the die.
Give a reason for your answer. (2)



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Question 7 continued

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Question 7 continued

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Question 7 continued

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Question 7 continued

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(Total for Question 7 is 15 marks)

TOTAL FOR PAPER IS 75 MARKS



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Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Tuesday 18 June 2019			
Morning (Time: 1 hour 30 minutes)		Paper Reference 9FM0/3B	
Further Mathematics			
Advanced			
Paper 3B: Further Statistics 1			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

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- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
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Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

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- (2)

(3)

- (1)

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. Indre works on reception in an office and deals with all the telephone calls that arrive. Calls arrive randomly and, in a 4-hour morning shift, there are on average 80 calls.

- (a) Using a suitable model, find the probability of more than 4 calls arriving in a particular 20-minute period one morning.

(3)

Indre is allowed 20 minutes of break time during each 4-hour morning shift, which she can take in 5-minute periods. When she takes a break, a machine records details of any call in the office that Indre has missed.

One morning Indre took her break time in 4 periods of 5 minutes each.

- (b) Find the probability that in exactly 3 of these periods there were no calls.

(2)

On another occasion Indre took 1 break of 5 minutes and 1 break of 15 minutes.

- (c) Find the probability that Indre missed exactly 1 call in each of these 2 breaks.

(3)



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Question 2 continued

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Question 2 continued

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Question 2 continued

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(Total for Question 2 is 8 marks)



3. A biased spinner can land on the numbers 1, 2, 3, 4 or 5 with the following probabilities.

Number on spinner	1	2	3	4	5
Probability	0.3	0.1	0.2	0.1	0.3

The spinner will be spun 80 times and the mean of the numbers it lands on will be calculated.
Find an estimate of the probability that this mean will be greater than 3.25

(6)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 6 marks)



4. Liam and Simone are studying the distribution of oak trees in some woodland. They divided the woodland into 80 equal squares and recorded the number of oak trees in each square. The results are summarised in Table 1 below.

Number of oak trees in a square	0	1	2	3	4	5	6	7 or more
Frequency	1	4	21	23	13	11	7	0

Table 1

Liam believes that the oak trees were deliberately planted, with 6 oak trees per square and that a constant proportion p of the oak trees survived.

- (a) Suggest the model Liam should use to describe the number of oak trees per square. (2)

Liam decides to test whether or not his model is suitable and calculates the expected frequencies given in Table 2.

Number of oak trees in a square	0 or 1	2	3	4	5	6
Expected frequency	5.53	14.89	24.26	22.24	10.87	2.21

Table 2

- (b) Showing your working clearly, complete the test using a 5% level of significance. You should state your critical value and conclusion clearly. (7)

Simone believes that a Poisson distribution could be used to model the number of oak trees per square. She calculates the expected frequencies given in Table 3.

Number of oak trees in a square	0 or 1	2	3	4	5	6 or more
Expected frequency	12.69	16.07	s	14.58	t	9.37

Table 3

- (c) Find the value of s and the value of t , giving your answers to 2 decimal places. (4)

- (d) Write down hypotheses to test the suitability of Simone's model. (1)

The test statistic for this test is 8.749

- (e) Complete the test. Use a 5% level of significance and state your critical value and conclusion clearly. (3)

- (f) Using the results of these tests, explain whether the origin of this woodland is likely to be cultivated or wild. (2)



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Question 4 continued

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Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 19 marks)



5. Information was collected about accidents on the *Seapron* bypass. It was found that the number of accidents per month could be modelled by a Poisson distribution with mean 2.5

Following some work on the bypass, the numbers of accidents during a series of 3-month periods were recorded. The data were used to test whether or not there was a change in the mean number of accidents per month.

- (a) Stating your hypotheses clearly and using a 5% level of significance, find the critical region for this test. You should state the probability in each tail. (5)
- (b) State $P(\text{Type I error})$ using this test. (1)

Data from the series of 3-month periods are recorded for 2 years.

- (c) Find the probability that at least 2 of these 3-month periods give a significant result. (3)

Given that the number of accidents per month on the bypass, after the work is completed, is actually 2.1 per month,

- (d) find $P(\text{Type II error})$ for the test in part (a)

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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 12 marks)



6. The discrete random variable X has probability generating function

$$G_X(t) = k \ln \left(\frac{2}{2-t} \right)$$

where k is a constant.

- (a) Find the exact value of k (1)
- (b) Find the exact value of $\text{Var}(X)$ (7)
- (c) Find $P(X = 3)$ (4)



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Question 6 continued

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Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 12 marks)



7. A spinner can land on red or blue. When the spinner is spun, there is a probability of $\frac{1}{3}$ that it lands on blue. The spinner is spun repeatedly.

The random variable B represents the number of the spin when the spinner first lands on blue.

(a) Find (i) $P(B = 4)$

(ii) $P(B \leq 5)$

(4)

(b) Find $E(B^2)$

(3)

Steve invites Tamara to play a game with this spinner.

Tamara must choose a colour, either red or blue.

Steve will spin the spinner repeatedly until the spinner first lands on the colour Tamara has chosen. The random variable X represents the number of the spin when this occurs.

If Tamara chooses red, her score is e^X

If Tamara chooses blue, her score is X^2

- (c) State, giving your reasons and showing any calculations you have made, which colour you would recommend that Tamara chooses.

(5)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Mock Paper Set 1			
(Time: 1 hour 30 minutes)		Paper Reference 9FM0/3B	
Further Mathematics Advanced Paper 3B: Further Statistics 1			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
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- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

1. In a survey, a large number of people were asked whether or not they recycle plastics. In each case the distance, in metres, from their home to the nearest plastic recycling point was measured and recorded. The results are shown in Table 1

Observed frequencies		Recycle plastics	
		Yes	No
Distance to nearest plastic recycling point	Less than 500 m	64	42
	500–1000 m	32	22
	More than 1000 m	14	26

Table 1

The Council's Environment Officer, Barbara, believes that whether or not people recycle plastics is independent of the distance to the nearest recycling point.

Barbara decides to test if the data from the survey supports her belief. Her expected frequencies are shown in Table 2

Expected frequencies		Recycle plastics	
		Yes	No
Distance to nearest plastic recycling point	Less than 500 m	58.3	47.7
	500–1000 m	29.7	24.3
	More than 1000 m	22	18

Table 2

- (a) Carry out a hypothesis test, at the 5% significance level, to see if the data from the survey support Barbara's belief. State your hypotheses and the critical value used in this test.

(5)

Barbara explains her results to her assistant, Bill. The data were collected by Bill, who now confesses to Barbara that he actually only collected data from 100 people and then doubled all his results.

- (b) With reference to the test statistic and critical value, explain whether or not Barbara's conclusions are still valid.

(2)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 7 marks)



2. A bag contains 10 tokens, 3 are red and 7 are green. Tokens are removed, one at a time, without replacement until a red token is removed or until 4 tokens have been removed, whichever occurs first.

The discrete random variable X represents the number of tokens removed.

(a) Show that $P(X = 4) = \frac{7}{24}$ (2)

The probability distribution of X is given in the table below

x	1	2	3	4
$P(X=x)$	$\frac{3}{10}$	$\frac{7}{30}$	$\frac{7}{40}$	$\frac{7}{24}$

(b) Showing your working clearly, find $\text{Var}(X)$ (4)

The same bag of 10 tokens is now used again. This time a token is removed, the colour is noted and the token is replaced. This procedure is repeated until a red token is removed.

The discrete random variable Y represents the number of green tokens removed.

(c) Showing your working clearly, find $\text{Var}(Y)$ (4)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 10 marks)



3. The discrete random variable X has distribution $X \sim \text{Po}(10.8)$

(a) (i) Find the probability that $X = 9$

(ii) Find the probability that X is at least 9

(3)

Charlie has a mobile phone. His parents claim that on average he looks at his mobile phone no more than 3 times every 30 minutes. Charlie's teacher suspects that this may not be true and that Charlie looks at his mobile phone more frequently than 3 times every 30 minutes. She decides to carry out a hypothesis test of her belief, at a 5% significance level.

She observes Charlie during a 1-hour revision class and notes the number of times that he looks at his mobile phone.

If Charlie actually looks at his mobile phone at an average rate of 5 times every 30 minutes,

(b) find the power of the teacher's hypothesis test, stating her hypotheses.

(6)

(c) State an assumption the teacher needs to make in order for this test to be valid.

(1)



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Question 3 continued

Lined area for writing the answer to Question 3.



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Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 10 marks)



4. Desmond believes that a particular six-sided die is biased. He decides to count the number of rolls required until he first rolls a 5 or a 6. He carries out this experiment 100 times. His results are summarised in the table below

Number of rolls to 1st 5 or 6	1	2	3	4	5	6	7 or more
Observed frequency	39	17	17	10	3	3	11

- (a) Stating your hypotheses clearly, test, at the 5% significance level, whether or not these results provide evidence that the die is biased. You should show your working clearly, including the expected frequencies and the critical value used. (10)

Mai suggests recording the results of rolling the die 100 times and using a goodness of fit test for a discrete uniform distribution.

- (b) (i) Give 2 reasons why Mai might think that her test is better than Desmond's test.
 (ii) Give a reason why Desmond might think that his test is better than Mai's test. (3)



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Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 13 marks)



5. A laboratory checks soil samples for evidence of lead contamination. Past records suggest that 0.5% of samples test positive for lead contamination.

As part of a new contract the laboratory receives batches of 1000 independent soil samples. The random variable X represents the number of soil samples that test positive for lead contamination out of a batch of 1000 independent soil samples.

Ushma wants to calculate the probability that at least 10 soil samples will test positive for lead contamination out of a batch of 1000 independent soil samples.

- (a) (i) State the distribution of X .
- (ii) Calculate the percentage error if Ushma uses a Poisson distribution to estimate this probability.
- (6)

Ushma decides to use $X \geq 9$ as the critical region to test the hypotheses

$$H_0: p = 0.005 \quad \text{against} \quad H_1: p > 0.005$$

where p is the probability that a randomly selected soil sample tests positive for lead contamination.

Ushma's colleague Javed suggests an alternative test of size 8%. Their manager Chris wants to minimise the probability of a Type I error. Given no further information about Javed's test,

- (b) explain which of these tests you would advise Chris to use. Give a reason for your answer.
- (4)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 10 marks)



6. Malik is a circus clown. As a part of his act he throws darts at a modified dartboard whilst blindfolded. He knows that there is a probability of 0.8 that each dart misses the bullseye.

In rehearsals, Malik throws darts at the dartboard whilst blindfolded until his assistant tells him he has hit the bullseye 3 times.

- (a) (i) Calculate how many darts Malik should expect to throw in order to hit the bullseye 3 times. (3)
- (ii) State two assumptions that need to be made about Malik's throws of the darts in order for this calculation to be valid. (2)

Malik rehearses once a day for each of 100 days.

Given that the assumptions in part (a) are true,

- (b) calculate an estimate of the probability that the mean number of throws required is at least 16 (5)



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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 10 marks)



7. A discrete random variable X has probability generating function $G_X(t) = e^{\lambda(t-1)}$ where λ is a positive constant.

(a) Use calculus to show that $E(X) = \lambda$

(3)

A discrete random variable Y has probability generating function $G_Y(t) = \left(\frac{t^2}{a} + b\right)^n$ where a and b are both positive and n is a positive integer.

(b) Show that

$$b = 1 - \frac{1}{a}$$

(2)

(c) Hence find the range of possible values of a

(2)

(d) In the case where $n = 4$, use calculus to show that $\text{Var}(Y) = \frac{16}{a^2}(a - 1)$

(8)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 15 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Mock Paper Set 2			
(Time: 1 hour 30 minutes)		Paper Reference 9FM0/3B	
Further Mathematics			
Advanced			
Paper 3B: Further Statistics 1			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Values from statistical tables should be quoted in full. If a calculator is used instead of the tables the value should be given to an equivalent degree of accuracy.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

1. The random variable X has probability distribution $X \sim B(120, 0.1)$

Show that use of the Poisson approximation to estimate $P(X < 10)$ gives an error of 0.0138 to 3 significant figures.

(4)

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



2. The discrete random variable X has probability generating function

$$G_X(t) = \frac{t^3}{k} \left((1+t)^2 + 2t^4 \right)$$

- (a) Find the value of the constant k . (2)

- (b) Showing your working, find $P(4 \leq X \leq 6.6)$

Given that X_1 and X_2 are independent observations of the random variable X

- (c) write down the probability generating function of the random variable $W = X_1 + X_2$ (1)



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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 8 marks)



3. Jack and Gill have an argument about which one of them is best at bringing back buckets of water from the well at the top of the hill. Jack claims that he can successfully bring back more buckets of water from the well in one hour than Gill can.

Jack consistently makes 3 journeys to the well and back each hour and the journeys are independent of each other. On each journey the probability that he is unsuccessful in bringing back a bucket of water is 0.4, because he falls over and loses the water.

The number of successful journeys that Gill can make in one hour, Y , is modelled by a Poisson distribution with mean 2.1

The number of successful journeys that Jack makes is independent of the number of successful journeys that Gill makes.

Showing your working, calculate the probability that in one hour,

- (a) Jack and Gill each successfully bring back 3 buckets of water from the well, (3)
- (b) Jack successfully brings back more buckets of water from the well than Gill. (4)
- (c) Explain why the Poisson distribution may not be a reasonable model for the number of successful journeys that Gill can make in one hour. (1)



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Question 3 continued

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Question 3 continued

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 8 marks)



4. Bob's hobby is using his metal detector to look for objects in the ground. The local farmer, Diane, lets him look in a large field. When the metal detector signals a find to Bob he digs it up and classifies it as either interesting or rubbish. Only 10% of finds are classified as interesting.

Bob looks for objects in the large field every day for 5 consecutive days. He decides that he will keep going each day until he has found one interesting object and then he will stop.

- (a) State an assumption Bob must make to model the number of objects he digs up in a day with a geometric distribution.

(1)

Using this assumption, calculate the probability that

- (b) he digs up 7 objects on the first day,

(3)

- (c) he digs up more than 7 objects on the first day,

(3)

- (d) he digs up fewer than 7 objects each day on 3 out of the 5 days.

(3)

Over the summer Bob looks for objects in the large field for a total of 80 days.

- (e) Estimate the probability that the mean number of objects which Bob digs up each day is fewer than 9

(4)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 14 marks)



5. A robot is programmed to perform a random walk along a number line.

In each minute it moves either +1 along the number line with probability $\frac{2}{3}$, or -1 along the number line with probability $\frac{1}{3}$, each move being independent of all previous moves.

Given that the robot starts at position 0 and that the random variable Y represents its position after 3 moves,

(a) show that $P(Y = 1) = \frac{4}{9}$ (2)

(b) showing each stage of your working, calculate the variance of Y (6)

If the random variable D represents its distance from position 0 after 3 moves,

(c) explain whether $\text{Var}(D)$ is the same, greater than or less than $\text{Var}(Y)$. (3)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 11 marks)



6. At a seaside town, one tourist attraction is a whale watching boat trip. The owner claims that on afternoon trips, which each have a duration of four hours, there have been an average of 2.4 whale sightings per trip.

A rival business doubts this claim and sends some employees to monitor whale sightings on 60 afternoon trips. The results are shown in the table below.

Number of sightings	0	1	2	3	4	5	6 or more
Number of trips	4	18	20	8	6	3	1

The number of whale sightings per trip is modelled by a Poisson distribution.

- (a) Test the owner's claim at the 10% significance level. Show your working, including the expected frequencies, the test statistic and the critical value, and state your hypotheses clearly.

(9)

During one of the trips the only sighting was a single group of 8 whales.

- (b) With reference to the model, state giving a reason, how this sighting should have been recorded.

(2)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 11 marks)



7. The random variable $X \sim \text{Geo}(p)$

A test is to be carried out using the hypotheses

$$H_0: p = 0.4 \qquad H_1: p < 0.4$$

Given that the probability of a Type I error is to be as close as possible to 5%

(a) find the critical region for the test. Show your working clearly.

(6)

Given that actually $p = \beta$ and that the probability of a Type II error for this test is 0.5

(b) find the value of β giving your answer to 3 significant figures.

(4)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 10 marks)



8. Jo and Keith are carrying out an experiment. A bag contains a large number of counters, some of which are green. Keith randomly draws a counter from the bag, records its colour and then replaces it before drawing the next counter. He continues until he has drawn two green counters.

The random variable X represents the number of draws up to and including the second green counter.

The random variable Y represents the number of draws up to and including the first green counter.

On each occasion Jo records the value of x and the value of y . They repeat this process a large number of times.

- (a) State an appropriate distribution to model the random variable Y (1)

Looking at the data, Jo suggests that the variance of X might be equal to the square of the mean of X

- (b) Given that X may be modelled by the negative binomial distribution, explain why Jo's suggestion cannot be true for the underlying distribution. (3)

Jo uses the data and estimates that the variance of Y is 20

- (c) Calculate an estimate of the proportion of green counters in the bag. (3)

In a different bag 40% of the counters are green and the rest are red.

- (d) For this bag, calculate the probability that the second green counter occurs on the 5th draw, given that the first green counter occurred on the 2nd draw. (2)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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(Total for Question 8 is 9 marks)

TOTAL FOR PAPER IS 75 MARKS



Write your name here

Surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Further Mathematics

Advanced

Further Mathematics Option 1

Paper 3: Further Statistics 1

Further Mathematics Option 2

Paper 4: Further Statistics 1

Sample Assessment Material for first teaching September 2017

Time: 1 hour 30 minutes

Paper Reference

9FM0/3B

9FM0/4B

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

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- Answers should be given to three significant figures unless otherwise stated.

Information

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- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

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- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

1. Bacteria are randomly distributed in a river at a rate of 5 per litre of water. A new factory opens and a scientist claims it is polluting the river with bacteria. He takes a sample of 0.5 litres of water from the river near the factory and finds that it contains 7 bacteria. Stating your hypotheses clearly test, at the 5% level of significance, whether there is evidence that the level of pollution has increased.

(5)

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2. A call centre routes incoming telephone calls to agents who have specialist knowledge to deal with the call. The probability of a caller, chosen at random, being connected to the wrong agent is p .

The probability of at least 1 call in 5 consecutive calls being connected to the wrong agent is 0.049

The call centre receives 1000 calls each day.

- (a) Find the mean and variance of the number of wrongly connected calls a day. (7)

- (b) Use a Poisson approximation to find, to 3 decimal places, the probability that more than 6 calls each day are connected to the wrong agent.

- (c) Explain why the approximation used in part (b) is valid. (2)

The probability that more than 6 calls each day are connected to the wrong agent using the binomial distribution is 0.8711 to 4 decimal places.

- (d) Comment on the accuracy of your answer in part (b). (1)

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Question 2 continued

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Pearson Edexcel Level 3 Advanced GCE in Mathematics
Sample Assessment Materials – Issue 1 – June 2017 © Pearson Education Limited 2017

3. Bags of £1 coins are paid into a bank. Each bag contains 20 coins.

The bank manager believes that 5% of the £1 coins paid into the bank are fakes. He decides to use the distribution $X \sim B(20, 0.05)$ to model the random variable X , the number of fake £1 coins in each bag.

The bank manager checks a random sample of 150 bags of £1 coins and records the number of fake coins found in each bag. His results are summarised in Table 1. He then calculates some of the expected frequencies, correct to 1 decimal place.

Number of fake coins in each bag	0	1	2	3	4 or more
Observed frequency	43	62	26	13	6
Expected frequency	53.8	56.6		8.9	

Table 1

- (a) Carry out a hypothesis test, at the 5% significance level, to see if the data supports the bank manager's statistical model. State your hypotheses clearly.

(10)

The assistant manager thinks that a binomial distribution is a good model but suggests that the proportion of fake coins is higher than 5%. She calculates the actual proportion of fake coins in the sample and uses this value to carry out a new hypothesis test on the data. Her expected frequencies are shown in Table 2.

Number of fake coins in each bag	0	1	2	3	4 or more
Observed frequency	43	62	26	13	6
Expected frequency	44.5	55.7	33.2	12.5	4.1

Table 2

- (b) Explain why there are 2 degrees of freedom in this case.
- (c) Given that she obtains a χ^2 test statistic of 2.67, test the assistant manager's hypothesis that the binomial distribution is a good model for the number of fake coins in each bag. Use a 5% level of significance and state your hypotheses clearly.

(2)

(2)

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Question 3 continued

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Question 3 continued

(Total for Question 3 is 14 marks)

4. A random sample of 100 observations is taken from a Poisson distribution with mean 2.3

Estimate the probability that the mean of the sample is greater than 2.5

(4)

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5. The probability of Richard winning a prize in a game at the fair is 0.15

Richard plays a number of games.

- (a) Find the probability of Richard winning his second prize on his 8th game, (2)

- (b) State two assumptions that have to be made, for the model used in part (a) to be valid. (2)

Mary plays the same game, but has a different probability of winning a prize. She plays until she has won r prizes. The random variable G represents the total number of games Mary plays.

- (c) Given that the mean and standard deviation of G are 18 and 6 respectively, determine whether Richard or Mary has the greater probability of winning a prize in a game. (4)

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6. The probability generating function of the discrete random variable X is given by

$$G_X(t) = k(3 + t + 2t^2)^2$$

(a) Show that $k = \frac{1}{36}$ (2)

(b) Find $P(X = 3)$ (2)

(c) Show that $\text{Var}(X) = \frac{29}{18}$ (8)

(d) Find the probability generating function of $2X + 1$ (2)

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Question 6 continued

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7. Sam and Tessa are testing a spinner to see if the probability, p , of it landing on red is less than $\frac{1}{5}$. They both use a 10% significance level.

Sam decides to spin the spinner 20 times and record the number of times it lands on red.

- (a) Find the critical region for Sam's test. (2)

- (b) Write down the size of Sam's test. (1)

Tessa decides to spin the spinner until it lands on red and she records the number of spins.

- (c) Find the critical region for Tessa's test. (6)

- (d) Find the size of Tessa's test. (1)

- (e) (i) Show that the power function for Sam's test is given by

$$(1 - p)^{19}(1 + 19p)$$

- (ii) Find the power function for Tessa's test. (4)

- (f) With reference to parts (b), (d) and (e), state, giving your reasons, whether you would recommend Sam's test or Tessa's test when $p = 0.15$ (4)

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Question 7 continued

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Question 7 continued

(Total for Question 7 is 18 marks)

TOTAL FOR PAPER IS 75 MARKS

Please check the examination details below before entering your candidate information

Candidate surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Specimen Paper

(Time: 1 hour 30 minutes)

Paper Reference **9FM0/3B**

Further Mathematics
Advanced
Paper 3B: Further Statistics 1

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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(8)

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 8 marks)



2. The discrete random variable Y has probability distribution given by

y	a	2	7
$P(Y=y)$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{5}$

where a is a constant.

Given that $\text{Var}(Y) = 28$ and $E(Y) < 0$

(a) find the value of a .

(6)

(b) Find $E\left(\frac{1}{3-Y}\right)$

(2)



Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 8 marks)



3. Tim and Sue are each typing a manuscript and they make errors at random.

On average, Tim makes 0.45 errors per page and Sue makes 0.2 errors per page.

A random sample of n pages of Tim's typing is taken. The probability that these n pages contain no errors is less than 0.05

- (a) Find the smallest possible value of n .

(3)

The random variable X represents the total number of errors in a random sample of 5 pages of Tim's typing and 5 pages of Sue's typing.

- (b) Find $P(X = 2)$, stating a necessary assumption.

(3)

Random samples, each consisting of 5 pages of Tim's typing and 5 pages of Sue's typing, are selected.

- (c) Find the probability that in 10 of these samples there are at least 2 with no errors.

(4)



Question 3 continued

Handwriting practice area with 30 horizontal lines.

(Total for Question 3 is 10 marks)



4. The discrete random variable $X \sim B(n, p)$ has probability generating function given by

$$G_x(t) = (0.4 + 0.6t)^2$$

- (a) Write down the value of n and the value of p (2)

Using the probability generating function, find

- $$(b) \text{ (i) } P(X = 1) \tag{2}$$

- $$(ii) \quad E(X) \tag{3}$$

Two independent observations, X_1 and X_2 , are taken from the distribution of X . The random variable $Y = X_1 + X_2$

- (c) Use calculus to show that $E(Y^2) = 6.72$ (7)

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 14 marks)



5. A company claims that the proportion of visitors to its website who make a purchase is 0.03

Nina believes that the proportion of visitors to the website who make a purchase is less than 0.03 and asks the company for some data to test this.

The company tells Nina, for a particular day, the number of visits to its website until a purchase is made. Nina assumes that visits to the website are made independently and that each visitor has the same probability of making a purchase.

- (a) Using a 5% level of significance, find the critical region for this test. State your hypotheses clearly.

(5)

- (b) Find the actual level of significance for this test.

(2)

The 94th visitor to the website was the first person to make a purchase.

- (c) Test, at the 5% level of significance, whether or not there is evidence that the proportion of visitors who make a purchase is less than 0.03.

(2)

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Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

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Question 5 continued

Handwriting practice area with 30 horizontal lines.

(Total for Question 5 is 9 marks)



6. The weights of bars of soap are known to be normally distributed.

The standard deviation of the weights is 3 grams.

A label on the bars of soap states that the mean weight is 120 grams.

Gizel believes that the mean weight of the bars of soap is greater than 120 grams. She takes a random sample of 10 bars of soap and finds the mean weight of her sample.

Gizel then tests, at the 5% level of significance, whether there is evidence that the bars of soap weigh more than 120 grams.

- (a) Write down the probability of a Type I error. (1)

Given that the true mean weight of the bars of soap is 122 grams,

- (b) show that the power of Gisel's test is 0.68 to 2 significant figures. (5)

Alex decides to carry out the same test at the 1% level of significance.

- (c) Without carrying out the test, compare the power of Alex's test with the power of Gizel's test. Give a reason for your answer. (2)

Joseph decides to increase the sample size and carry out the test at the 5% level of significance.

Assuming that the true mean weight of the bars of soap is 122 grams,

- (d) calculate the smallest sample size that will produce a power of at least 0.9

- (e) State which of Gizel's and Joseph's tests should be used.
Give a reason for your answer.
- (2)**



Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 15 marks)



7. A radio station is running a contest each day for 30 days.

Each day it awards a prize to each of the first 12 callers who phone in and answer a question correctly. Once 12 prizes are awarded, no more calls are taken that day.

It can be assumed that each caller has a $\frac{3}{4}$ chance of answering a question correctly, independently of all other callers.

- (a) Find the probability that there are exactly 15 calls taken on the first day. (2)
- (b) Find the probability that there are more than 13 calls taken on the first day. (3)
- (c) Estimate the probability that the mean number of calls taken over the 30 days is more than 15.5 (6)

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Question 7 continued

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Question 7 continued

Lined area for writing answers.

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(Total for Question 7 is 11 marks)

TOTAL FOR PAPER IS 75 MARKS

